

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

Paper 1

Monday 15 May 2023

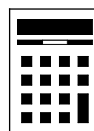
07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



Answer **all** questions in the spaces provided.

0 1

This question is about alkenes.

0 1 . 1

Hydrocarbons can be broken down to produce a mixture of alkanes and alkenes.

What is the name of this process?

[1 mark]

Tick (✓) **one** box.

Cracking

☐

Distillation

☐

Fermentation

☐

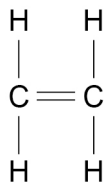
Precipitation

☐

Ethene is an alkene.

Figure 1 shows the displayed structure for ethene.

Figure 1



0 1 . 2 What is the formula of ethene?

[1 mark]

Tick (✓) **one** box.

CH₄

☐

C₂H

☐

C₂H₄

☐

C₄H₂

☐

0 1 . 3 What does = represent in **Figure 1**?

[1 mark]

Question 1 continues on the next page

Turn over ►



0 1 . 4 Describe the test for an alkene.

Give the result if an alkene is present.

[2 marks]

Test _____

Result _____

0 1 . 5 Ethene is used to produce a polymer.

What is the name of the polymer?

[1 mark]

0 1 . 6 Some used polymers are disposed of in landfill.

Complete the sentence.

[1 mark]

Disposing of polymers in landfill is a problem because many types of polymer
are not _____ .

0 1 . 7 Suggest **one** way to reduce the quantity of polymers disposed of in landfill.

[1 mark]



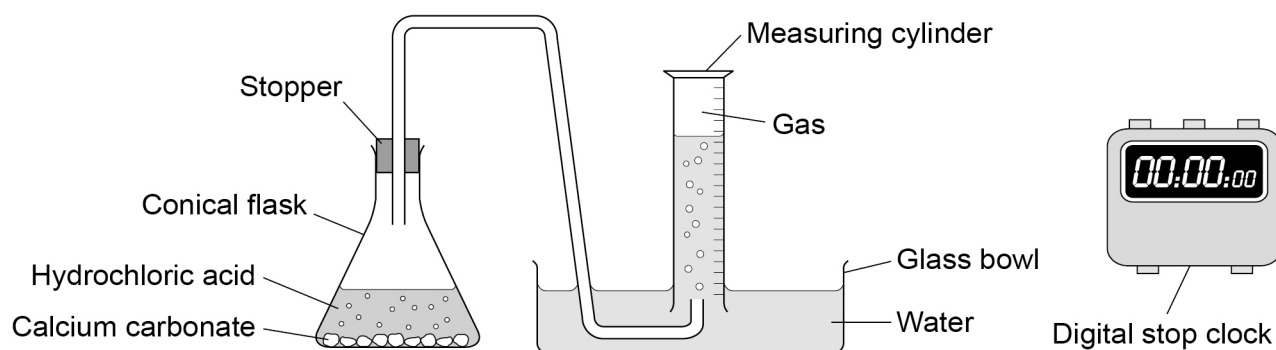
0 2

A student reacted calcium carbonate with hydrochloric acid.

The student investigated the effect of changing the size of calcium carbonate pieces on the rate of the reaction.

Figure 2 shows the apparatus used.

Figure 2



This is the method used.

- 1 Add 25 cm³ of hydrochloric acid to a conical flask.
- 2 Add 2.0 g of large pieces of calcium carbonate to the hydrochloric acid.
- 3 Insert the stopper and start the stop clock.
- 4 Record the volume of gas collected every 20 seconds for 300 seconds.
- 5 Repeat steps 1 to 4 using smaller pieces of calcium carbonate.

0 2 . 1

Which variable should the student keep the same to make the investigation a fair test?
[1 mark]

Tick (✓) **one** box.

Concentration of hydrochloric acid

☐

Size of calcium carbonate pieces

☐

Volume of gas collected

☐

Question 2 continues on the next page

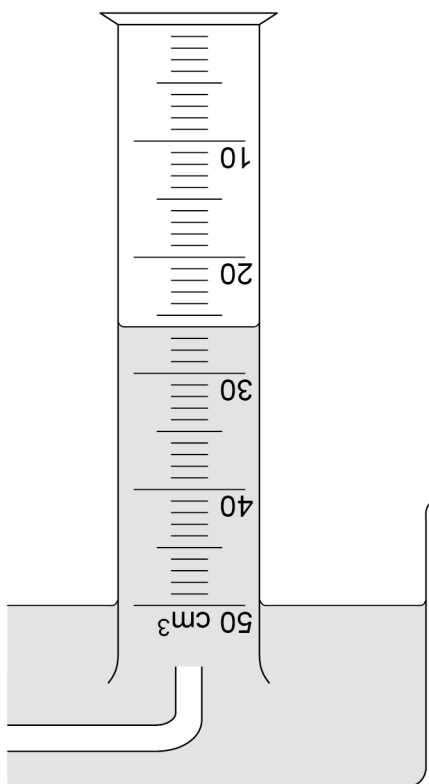
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0 2 . 2

Figure 3 shows the volume of gas collected in the measuring cylinder after 80 seconds using large pieces of calcium carbonate.

Figure 3



What is the volume of gas in the measuring cylinder?

[1 mark]

Volume = _____ cm³

0 2 . 3

The reaction using large pieces of calcium carbonate produced 43 cm³ of gas in 200 seconds.

Calculate the rate of reaction for large pieces of calcium carbonate.

Use the equation:

$$\text{rate of reaction} = \frac{\text{volume of gas produced}}{\text{time}}$$

[2 marks]

Rate of reaction = _____ cm³/s



The reaction using small pieces of calcium carbonate has a faster rate of reaction than the reaction using larger pieces of calcium carbonate.

0 2 . 4

Which **two** statements are correct for the reaction using the same mass of smaller pieces of calcium carbonate?

[2 marks]

Tick (✓) **two** boxes.

Smaller pieces of calcium carbonate have a greater surface area.

☐

Smaller pieces of calcium carbonate have more energy.

☐

The hydrochloric acid particles are moving faster.

☐

There are a greater number of particles per unit volume.

☐

There are more particle collisions per second.

☐

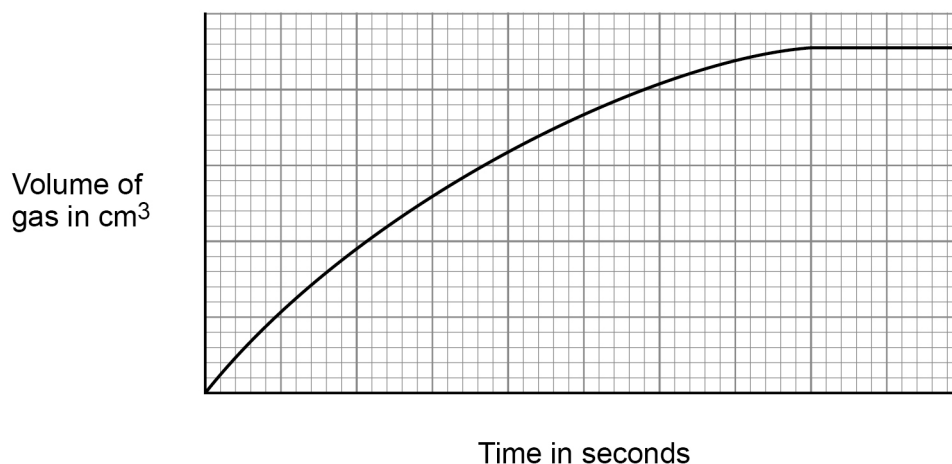
Question 2 continues on the next page

Turn over ►



0 2 . 5 **Figure 4** shows a sketch graph of the results for large pieces of calcium carbonate.

Figure 4



The experiment was repeated using the same mass of small pieces of calcium carbonate.

Draw a line on **Figure 4** to show the results for small pieces of calcium carbonate.

[2 marks]

0 2 . 6 Give **one** other method of changing the rate of reaction between calcium carbonate and hydrochloric acid.

You should **not** refer to the size of calcium carbonate pieces.

[1 mark]



0 3

This question is about endothermic energy changes.

0 3 . 1

Which of the following is a use of an endothermic energy change?

[1 mark]Tick (✓) **one** box.

Hand warmers

☐

Self-heating cans

☐

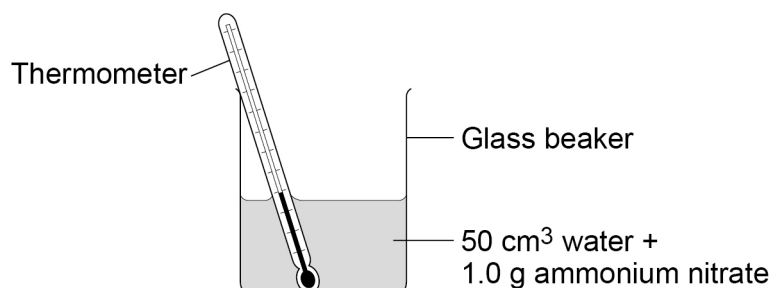
Sports injury packs

☐**Question 3 continues on the next page****Turn over ►**

A student investigated the effect of adding different masses of ammonium nitrate to water.

Figure 5 shows the apparatus.

Figure 5

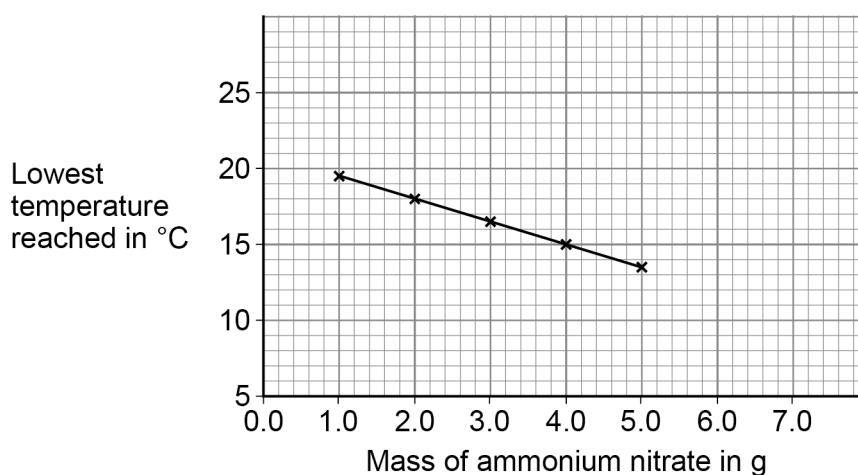


This is the method used.

- 1 Measure 50 cm³ of water into a glass beaker.
- 2 Add 1.0 g of ammonium nitrate to the water.
- 3 Stir until all the ammonium nitrate has dissolved.
- 4 Record the lowest temperature reached.
- 5 Repeat steps 1 to 4 with different masses of ammonium nitrate.

Figure 6 shows the results.

Figure 6



0 3 . 2

What happens to the lowest temperature reached when increasing masses of ammonium nitrate are added to 50 cm³ of water?

Use **Figure 6**.

[1 mark]

0 3 . 3

6.0 g of ammonium nitrate was added to 50 cm³ of water.

Suggest the lowest temperature reached.

You should extend the line of best fit on **Figure 6**.

[2 marks]

Temperature = _____ °C

0 3 . 4

Energy is transferred from the surroundings when ammonium nitrate dissolves in water.

Suggest **one** change to the apparatus in **Figure 5** that would decrease the rate of energy transfer.

[1 mark]

Question 3 continues on the next page

Turn over ►



0 3 . 5

The student repeated the investigation using ammonium chloride.

Table 1 shows the results for 5.0 g of ammonium chloride.

Table 1

Temperature of water before adding ammonium chloride in °C	18.5
Temperature of water after adding ammonium chloride in °C	13.0

Calculate the energy change when 5.0 g of ammonium chloride is added to 50 g of water.

Use the equation:

$$Q = mc\Delta T$$

Where:

- Q = energy change in J
- m = mass of water = 50 g
- c = specific heat capacity of water = 4.2 J/g °C
- ΔT = temperature change in °C.

[3 marks]

Energy change = _____ J

8



0 4

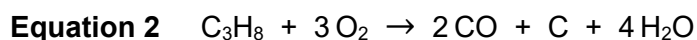
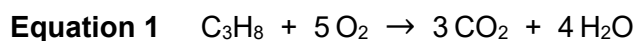
This question is about fuels.

Combustion of some fuels produces pollution, which affects the environment.

0 4 . 1

Propane gas is used as a fuel.

Two equations for the combustion of propane gas are:

Explain why the products in **Equation 2** are different from the products in **Equation 1**.
[2 marks]

0 4 . 2

Some fuels release solid particles (particulates) during combustion.

What environmental problem is caused by particulates?

[1 mark]

0 4 . 3

Combustion of some fuels causes acid rain.

Explain why.

[3 marks]

Question 4 continues on the next page

Turn over ►



0	4	.	4
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Table 2 compares diesel fuel and hydrogen fuel.

Table 2

	Diesel fuel	Hydrogen fuel
Source of fuel	Fractional distillation of crude oil	High temperature reaction between methane gas and steam
Products of reaction using fuel	Carbon dioxide and water	Water
State of fuel at room temperature and pressure	Liquid	Gas
Energy produced per gram of fuel in kJ	56	143

Evaluate the use of diesel fuel and hydrogen fuel to power cars.

Use Table 2.

[6 marks]

[illegible]

0 4 . 5 Diesel contains decane ($\text{C}_{10}\text{H}_{22}$).

19 500 J of energy is released from the combustion of 0.426 g of decane.

Calculate the energy change in kJ from the combustion of one mole of decane.

Relative atomic masses (A_r): H = 1 C = 12

[5 marks]

Energy change = _____ kJ

0 4 . 6 Combustion of a fuel is an exothermic reaction.

Explain exothermic reactions in terms of bond making and bond breaking.

[3 marks]

20

Turn over for the next question

Turn over ►



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0 5

This question is about zinc carbonate.

Zinc carbonate thermally decomposes when heated to produce zinc oxide and carbon dioxide.

0 5 . 1

Write a word equation for the thermal decomposition of zinc carbonate.

[1 mark]

0 5 . 2

Describe the test for carbon dioxide.

Give the result of the test if carbon dioxide is present.

[2 marks]Test

Result

0 5 . 3Calculate the percentage by mass of zinc in zinc carbonate (ZnCO_3).Relative atomic masses (A_r): C = 12 O = 16 Zn = 65**[3 marks]**

Percentage by mass = _____ %

Question 5 continues on the next page**Turn over ►**

A student investigated the thermal decomposition of zinc carbonate.

This is the method used.

- 1 Add zinc carbonate to a container.
- 2 Record the mass of the container and contents.
- 3 Heat the container and contents for 40 seconds.
- 4 Record the mass of the container and contents.
- 5 Repeat steps 3 and 4 for a total of 280 seconds.

Table 3 shows the results.

Table 3

Time in seconds	Mass of container and contents in grams
0	28.60
40	28.04
80	27.36
120	26.53
160	26.01
200	25.60
240	25.37
280	25.19

0 5 . 4

Calculate the mass of carbon dioxide released between 0 and 280 seconds.

[1 mark]

Mass of carbon dioxide released = _____ g



0 5 . 5

The student repeated steps **3** and **4** for a total of 600 seconds.

Suggest what would happen to the mass of the container and contents.

Give **one** reason for your answer.

[2 marks]

0 5 . 6

In the reaction, the zinc carbonate decomposed to produce 0.079 moles of carbon dioxide gas.

Calculate the volume of carbon dioxide gas produced in cm^3 .

The volume of 1 mole of carbon dioxide gas is 24 dm^3 .

Give your answer to 2 significant figures.

[4 marks]

Volume of carbon dioxide gas (2 significant figures) = _____ cm^3

13

Turn over for the next question

Turn over ►

0 6

This question is about transition metals.

0 6 . 1What is the name of the transition metal represented by ${}^{48}_{22}\text{Ti}$?

Use the periodic table.

[1 mark]

0 6 . 2Which **two** are properties of the transition metal copper?**[2 marks]**Tick (✓) **two** boxes.

Conducts heat

☐

Low melting point

☐

Reacts with water

☐

Soft

☐

Strong

☐**0 6 . 3**

Transition metals are elements.

What is meant by the word 'element'?

[1 mark]



Transition metals and their compounds have many uses.

0 6 . 4 Figure 7 shows a piece of pottery.

Figure 7



Transition metal compounds are used to improve the appearance of pottery.

Give **one** property of transition metal compounds that makes them suitable for this use.

[1 mark]

Transition metals can be used as catalysts.

0 6 . 5 Name the transition metal catalyst used in the Haber process.

[1 mark]

0 6 . 6 Explain how catalysts increase the rate of reactions.

[2 marks]

Question 6 continues on the next page

Turn over ►



0 6 . 7

The transition metal cobalt can produce different compounds with chlorine.

These compounds are:

- Cobalt(II) chloride (CoCl_2)
- Cobalt(III) chloride (CoCl_3).

Give **one** reason why cobalt can produce different compounds with chlorine.

[1 mark]

0 6 . 8

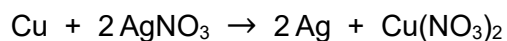
Describe the bonding in transition metals.

[3 marks]



Copper and silver are transition metals.

Silver is displaced from silver nitrate in a displacement reaction with copper.



0 6 . 9

Write the ionic equation for the displacement of silver by copper.

[2 marks]

0 6 . 1 0

Explain why the displacement of silver from silver nitrate is described as reduction.

[2 marks]

16

Turn over for the next question

Turn over ►



0 7

This question is about alcohols, carboxylic acids, and esters.

0 7 . 1

Alcohols contain the functional group –OH

What term is given to a series of compounds with the same functional group?

[1 mark]

0 7 . 2

Alcohols react with sodium to produce a gas.

What is the name of this gas?

[1 mark]

Tick (✓) **one** box.

Carbon dioxide

☐

Carbon monoxide

☐

Hydrogen

☐

Oxygen

☐

Ethanol is an alcohol.

Ethanol can be produced on a large scale in two ways:

- by fermenting sugar
- by reacting ethene with steam.

0 7 . 3 Ethanol dissolves in water to form a solution.

Give the pH of the solution.

[1 mark]

0 7 . 4 Write a balanced symbol equation for the production of ethanol by fermenting sugar ($\text{C}_6\text{H}_{12}\text{O}_6$).

[2 marks]

0 7 . 5 Give **two** advantages of using the reaction of ethene with steam compared with fermenting sugar to produce ethanol.

[2 marks]

1 _____

2 _____

0 7 . 6 Ethanoic acid can be produced from ethanol by microbial action.

What type of reaction is this?

[1 mark]

Question 7 continues on the next page

Turn over ►



0 7 . 7 Table 4 gives the displayed structures and boiling points of three alcohols.

Table 4

Name	Methanol	Ethanol	Propanol
Displayed structure	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{O}-\text{H} \\ \\ \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
Boiling Point in °C	65	78	97

Suggest why the boiling points of alcohols increase as the number of carbon atoms increase.

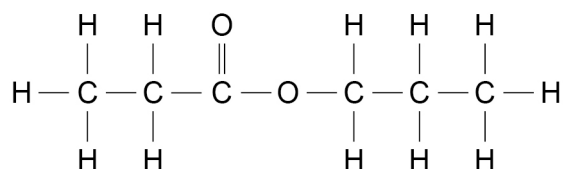
[3 marks]



Esters are produced when alcohols react with carboxylic acids.

Figure 8 shows the displayed structure of propyl propanoate.

Figure 8



0 7 . 8 Draw a circle around the ester functional group on **Figure 8**.

[1 mark]

0 7 . 9 Name the alcohol and the carboxylic acid used to produce propyl propanoate.

[2 marks]

Alcohol _____

Carboxylic acid _____

0 7 . 1 0 Esters are used in perfumes.

Give **two** properties that make esters suitable for use in perfumes.

[2 marks]

1 _____

2 _____

16

END OF QUESTIONS



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